

Code Optimization



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Definition

Code Optimization is a technique which tries to improve the code by eliminating unnecessary code lines and arranging the statements in such a sequence that speed up the program execution without wasting the resources.



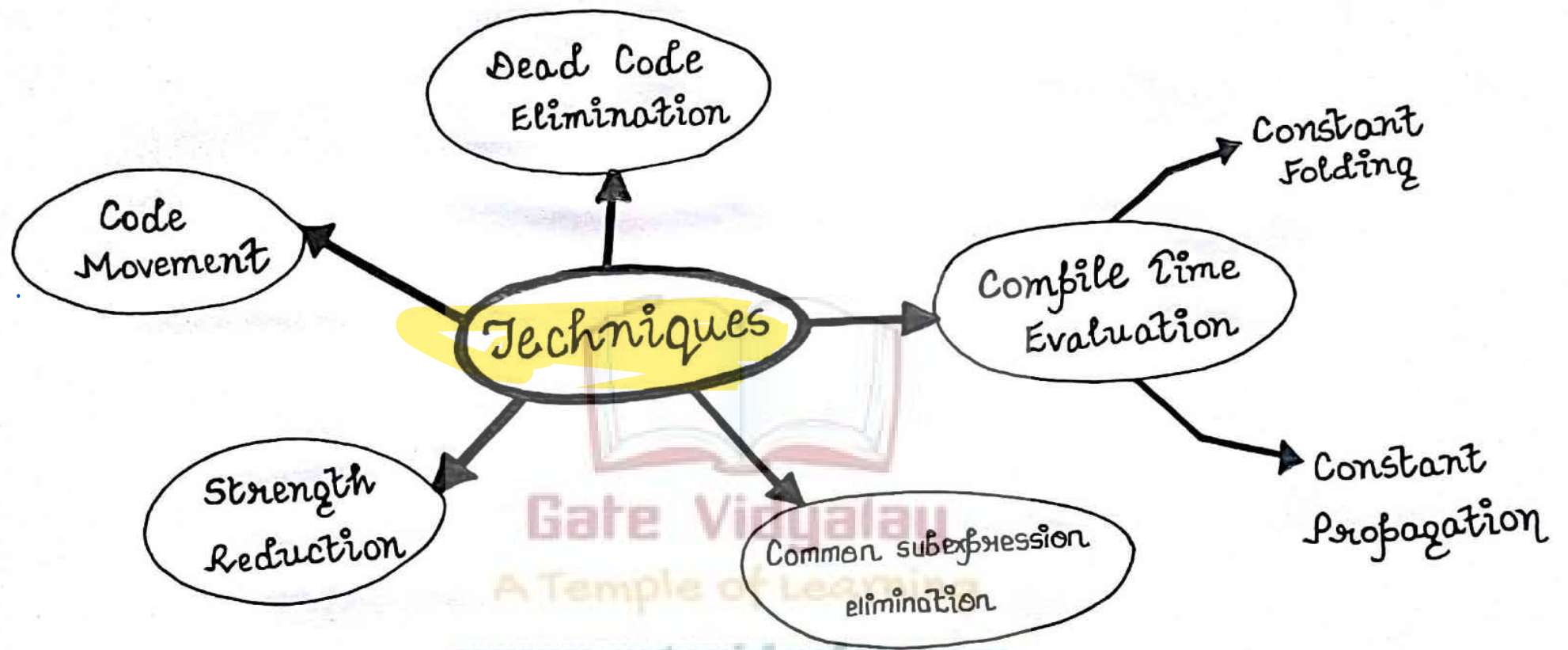
Advantages:-

- Executes faster
 - Efficient memory usage
 - Yields better performance
-

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① Compile Time Evaluation

i) Constant Folding:

It refers to a technique of evaluating the expressions whose operands are known to be constant at compile time itself.

Example:- $\text{length} = (2217) * d$

ii) Constant Propagation:

In constant propagation, if a variable is assigned a constant value, then subsequent use of that variable can be replaced by a constant as long as no intervening assignment has changed the value of the variable.

Example:-

$$\pi = 3.14$$

$$r = 5$$

$$\text{Area} = \pi * r * r$$

Here, the value of π is replaced by 3.14 and r by 5, then computation of $3.14 * r * r$ is done during compilation.

① Common sub-expression elimination:-

The common sub-expression is an expression appearing repeatedly in the code which is computed previously. This technique replaces redundant expression each time it is encountered.

Example:-

$$T1 = 4 * i$$

$$T2 = a[T1]$$

$$T3 = 4 * j$$

$$T4 = 4 * i$$

$$T5 = n$$

$$T6 = b[T4] + T5$$

Before Optimization

$$T1 = 4 * i$$

$$T2 = a[T1]$$

$$T3 = 4 * j$$

$$T5 = n$$

$$T6 = b[T4] + T5$$

After Optimization

III Code Movement:-

It is a technique of moving a block of code outside a loop if it won't have any difference if it is executed outside or inside the loop.

Example:-

$y=5$
 $z=6$

```
for (int i=0; i<n; i++)  
{  
    x = y+z;  
    a[i] = 6*i;  
}
```

Before Optimization

$y=5$
 $z=6$

```
x = y+z;  
for (int i=0; i<n; i++)  
{  
    a[i] = 6*i;  
}
```

After Optimization

④ Dead Code Elimination:-

Dead Code Elimination includes eliminating those code statements which are either never executed or unreachable or if executed their output is never used.

Example:-

```
i = 0  
if (i == 1)  
{  
    a = x + 5;  
}
```

Before Optimization



After Optimization

⑤ Strength Reduction:

It is the replacement of expressions that are expensive with cheaper and simple ones.

Example:

$$B = A * 2$$

Before Optimization

$$B = A + A$$

After Optimization

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